Threat Hunting Methodologies:

1. Introduction-

Threat hunting is a **proactive cybersecurity practice** that involves actively searching for hidden threats inside a network before they cause damage. Unlike traditional security methods that depend on alerts, threat hunting assumes that attackers may already be present and tries to detect them early.

It helps organizations-

- Detect advanced persistent threats (APTs)
- Reduce attacker dwell time
- Improve overall security posture

2. Core Concepts-

a) Proactive Threat Hunting- Proactive threat hunting is based on the idea of "assume breach".

Instead of waiting for alerts:

- Analysts create hypotheses based on attacker behavior.
- They investigate logs and system data to validate those hypotheses.

Types :-

- **Hypothesis-driven hunting** – Based on known attacker techniques (TTPs)
- **Data-driven hunting** – Based on anomalies in logs
- **Intelligence-driven hunting** – Based on threat intelligence feeds

b) Hunting Frameworks-

❖ SqRR Framework:

It stands for-

- **Search** → Identify suspicious indicators
- **Query** → Use queries on logs/data sources

- **Retrieve** → Extract relevant data
- **Respond** → Take action (block, isolate, alert)

❖ TaHiTi Framework:

It stands for-

- Targeted Hunting Integrating Threat Intelligence
- Combines threat intelligence with hunting activities
- Focuses on **specific adversaries and attack patterns**

c) Data Sources for Threat Hunting-

Effective hunting depends on analysing multiple data sources:

- **EDR Logs** → Endpoint activities (processes, files)
- **Network Traffic** → Suspicious connections, C2 traffic
- **SIEM Logs** → Aggregated logs from multiple systems
- **Threat Intelligence Feeds** → Known malicious IPs/domains
- **Authentication Logs** → Login anomalies, privilege misuse

3. Key Objectives of Threat Hunting-

- Identify threats before alerts are triggered
- Detect unknown and advanced attacks
- Reduce **Mean Time to Detect (MTTD)**
- Strengthen SOC capabilities
- Improve incident response readiness

Advanced SOAR Automation

1. Introduction-

SOAR stands for **Security Orchestration, Automation, and Response**. It is a cybersecurity approach that enables organizations to **automate repetitive security tasks**, integrate multiple tools, and respond to incidents efficiently.

SOAR platforms help Security Operations Centers (SOC) to:

- Reduce manual workload
- Improve response time
- Ensure consistent incident handling

2. Core Concepts-

a) Soar Components:

❖ Orchestration-

- Integrates different security tools (SIEM, EDR, firewalls).
- Enables smooth communication between systems.

❖ Automation-

- Automates repetitive and time-consuming tasks.
- Reduces human intervention.

❖ Response-

- Executes predefined actions to mitigate threats.

b) Playbook Development:

A **playbook** is a predefined workflow that defines how to handle specific security incidents.

c) Integration with SIEM and EDR:

SOAR works best when integrated with:

❖ SIEM

- Collects and correlates logs
- Sends alerts to SOAR

❖ EDR

- Provides endpoint-level visibility
- Enables automated response actions

3. Key Objectives of SOAR Automation

- Reduce **Mean Time to Respond (MTTR)**
- Automate repetitive SOC tasks
- Improve incident response efficiency

Post-Incident Analysis and Continuous Improvement

1. Introduction-
   Post-incident analysis is a critical phase of incident response that occurs **after a security incident has been contained and resolved**. It focuses on understanding what happened, why it happened, and how to prevent similar incidents in the future. This phase ensures **continuous improvement** of security operations and strengthens the organization's overall defense system.

2. Core Concepts-
   a) Root Cause Analysis (RCA):
      Root Cause Analysis (RCA) is used to identify the **actual cause of a security incident**, not just its symptoms.
      Techniques used in RCA-
      - ❖ 5 Whys Technique
        - o Repeatedly ask "Why?" (usually 5 times) to reach the root cause
      - ❖ Fishbone Diagram
        - o Visual method to identify multiple causes of a problem

   b) Lessons Learned Process:
      After resolving an incident, teams conduct a **post-mortem meeting** to evaluate performance.
      - ❖ Key Activities:
        - o Review timeline of the incident
        - o Identify what worked well
        - o Identify gaps and weaknesses
        - o Recommend improvements

3. Metrics and KPIs-
   Metrics help measure the effectiveness of SOC operations.
   Important Metrics are:

- **Mean Time to detect (MTTD)**-Time taken to identify a threat.
- **Mean Time to Respond (MTTR)**-Time taken to contain and mitigate a threat.
- **Incident Resolution Time**-Total time to fully resolve an incident.
- **False Positive Rate**- Number of incorrect alerts.

4. Key Objectives-
   - Identify root causes of incidents
   - Improve detection and response strategies
   - Reduce future risks
   - Strengthen SOC processes
   - Enhance team performance

Adversary Emulation Techniques

1. Introduction-

   Adversary emulation is a cybersecurity practice where defenders simulate real-world attacker behaviours to test and improve security systems. It helps organizations understand how attackers operate and whether their detection and response mechanisms are effective. Unlike penetration testing, adversary emulation focuses on **realistic attack scenarios based on known threat actors and their Tactics, Techniques, and Procedures (TTPs)**.

2. Core Concepts-

   a) Adversary Emulation

   Adversary emulation involves mimicking attacker activities using known techniques from frameworks like MITRE.

   - T1566- Phishing- Sending malicious email

b) Emulation Frameworks
- ❖ MITRE ATT&CK Framework
  - o A global knowledge base of attacker behaviours.
  - o Categorizes TTPs used by real attackers.
- ❖ MITRE Caldera
  - o An automated adversary emulation platform

c) Red Team vs Blue Team Collaboration
- ❖ Red Team- Simulates attacker
- ❖ Blue Team- Defends and detects threats

3. Key Objectives-
- Test real-world attack scenarios
- Identify weaknesses in security controls
- Improve detection and response capabilities
- Validate SOC readiness
- Enhance overall cybersecurity posture

Security Metrics and Executive Reporting

1. Introduction-

Security metrics and executive reporting are essential for evaluating the performance of a Security Operations Center (SOC) and communicating results to management. These metrics help organizations understand how effectively they are detecting, responding to, and preventing cyber threats. Executive reporting ensures that **non-technical stakeholders** can understand security performance and make informed decisions.

2. Core Concepts-

a) Advanced SOC Metrics- Security metrics are used to measure the effectiveness of security operations

Key Metrics

- ❖ Dwell Time
  - Time between initial compromise and detection
  - Lower dwell time= better security

b) Executive Reporting

Executive reporting focuses on presenting technical data in a **simple and understandable format**.

c) Continuous Improvement

Metrics are not just for reporting they are used to improve security.

3. Key Objectives-
   - Measure SOC performance
   - Improve detection and response efficiency
   - Support strategic decision-making
   - Communicate security posture to leadership
   - Drive continuous improvement

Threat Hunting Practice

1. Objective

   The objective of this task is to perform proactive threat hunting by developing a hypothesis, analysing logs, and validating findings using threat intelligence and endpoint data.

2. Tools Used
   - Elastic Security
   - Velociraptor
   - AlienVault OTX

3. Hypothesis Development

A threat hunting hypothesis was created to detect suspicious activity:

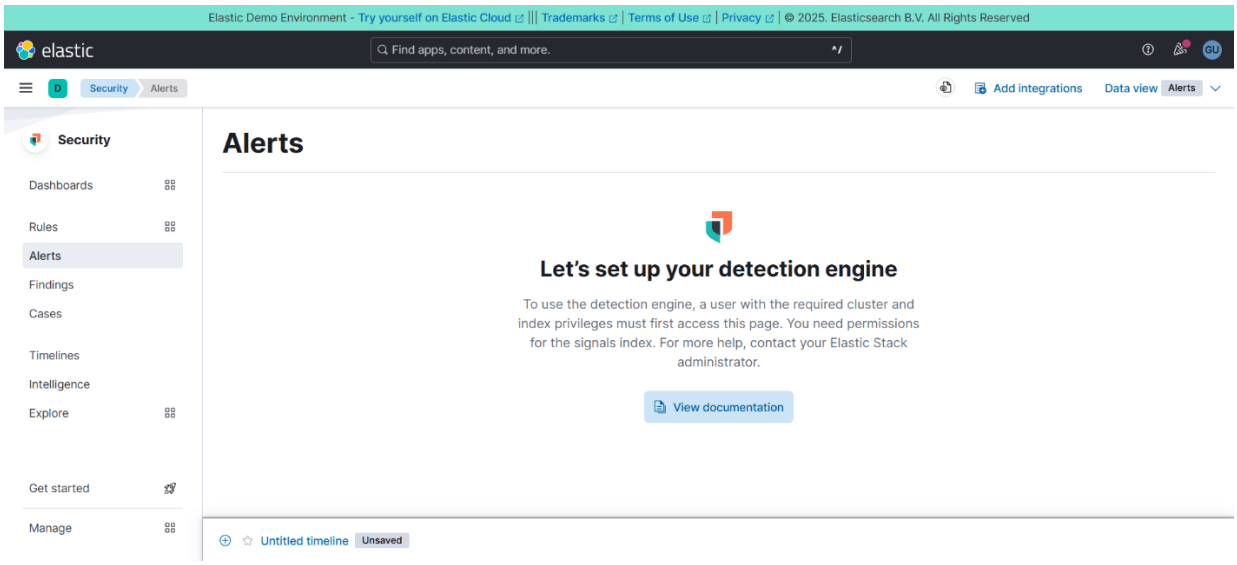
Hypothesis:

"Unauthorized privilege escalation may be occurring in domain accounts."

To validate this, logs were analysed for privileged access events such as **Event ID 4672**, which indicates special privileges assigned to a user.

4. Log Analysis using Elastic Security

The Elastic Security platform was used to analyse alerts and logs. The system required enabling the detection engine before proceeding with alert analysis.



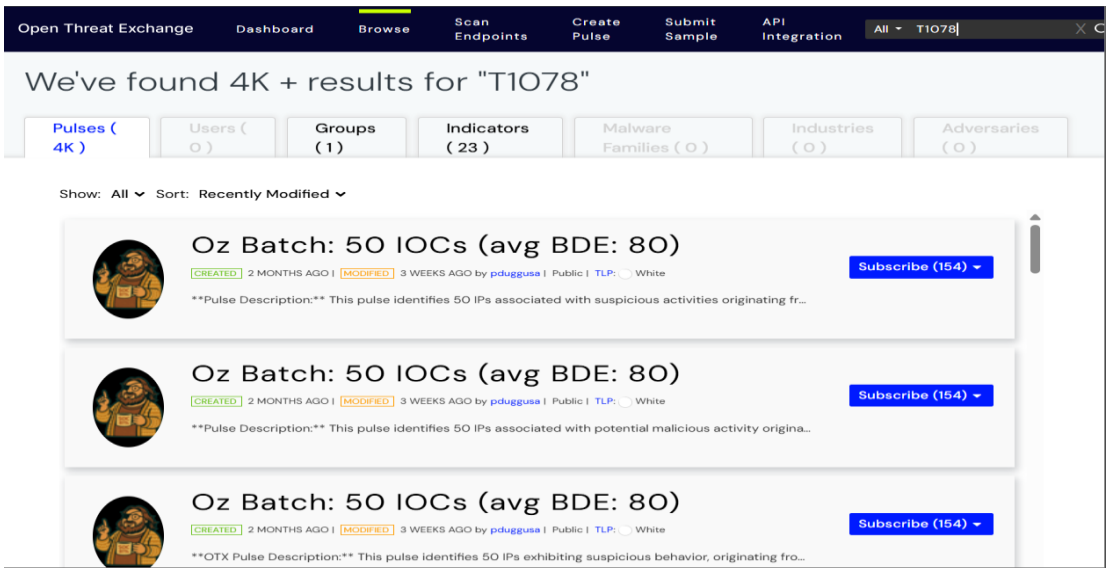
Elastic Security Alerts Dashboard

Elastic Security Alerts dashboard showing that the detection engine is not yet configured. This indicates that alert generation and threat detection cannot function until proper setup and permissions are enabled.

5. Threat Intelligence Hunting using AlienVault OTX-

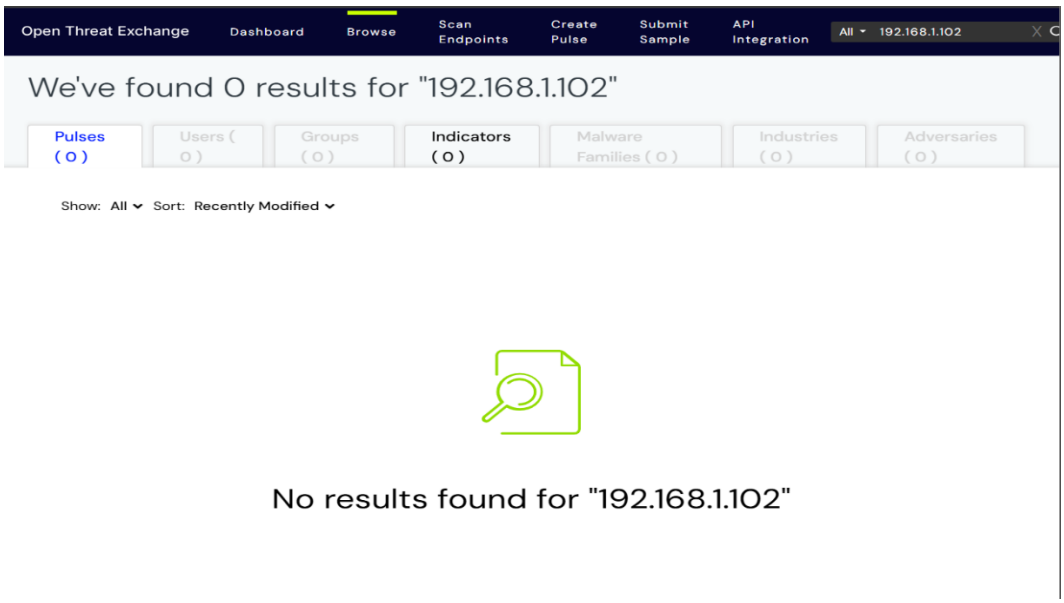
Threat intelligence was used to investigate MITRE ATT&CK technique:

T1078- Valid Accounts



OTX results for T1078 Account

- Further investigation was performed on specific IP 192.168.1.102



OTX result for IP 192.168.1.102

6. Endpoint Data Analysis using Velociraptor-

Endpoint investigation was performed using Velociraptor by querying process data.

```
SELECT * FROM processes WHERE username = 'testuser'
```

Velociraptor Query For Process Analysis of user 'testuser"

7. Finding Summary-

- No direct malicious indicators found in OTX for the tested IP.
- Threat intelligence revealed multiple indicators for T1078.
- Endpoint query helped in analysing user-level activity .
- Detection system setup is necessary for effective monitoring.

8. Hunting Report-

A threat hunting exercise was conducted based on the hypothesis of unauthorized privilege escalation using MITRE ATT&CK technique T1078 (Valid Accounts). Elastic Security was used to analyse logs, but detection required setup. Threat intelligence from AlienVault OTX provided multiple indicators related to T1078, although the tested IP (192.168.1.102) showed no public matches. Endpoint analysis using Velociraptor was performed to examine processes associated with the user "testuser." The investigation highlighted the importance of integrating threat intelligence and endpoint monitoring to detect suspicious account activity and improve proactive threat detection capabilities.

Furthermore, the absence of public threat intelligence for the tested IP suggests that internal monitoring and behavioural analysis are crucial for detecting potential insider or unknown threats.

SOAR Playbook Development

1. Objective
   The objective of this task is to design and implement an automated incident response playbook using SOAR tools to efficiently detect, analyse, and respond to phishing-related threats.

2. Tools Used
   - Splunk Phantom (SOAR)
   - TheHive
   - Wazuh
   - CrowdSec

3. Playbook Design
   A playbook was created to automate the response to phishing alerts.
   ❖ Steps In Playbook-
     - Detect phishing alert from SIEM (Wazuh)
     - Extract suspicious IP address
     - Check IP reputation
     - Block IP using CrowdSec
     - Create incident case in TheHive

4. Playbook Execution
   Detected IP: 192.168.1.102

Actions Performed:
- IP reputation checked → flagged as suspicious
- IP blocked using CrowdSec firewall
- Incident case created in TheHive

```
| Playbook Step   | Status   | Notes                                |
|-----------------|----------|--------------------------------------|
| Check IP        | Success  | IP flagged as malicious              |
| Block IP        | Success  | CrowdSec blocked 192.168.1.102       |
| Create Ticket   | Success  | Case created in TheHive              |
```

Playbook Execution

5. Automated Response

   The system automatically executed response actions:

   - IP blocking command executed
   - Threat contained successfully

```
Action: IP blocked using CrowdSec firewall

Command (simulated):
sudo cscli decisions add --ip 192.168.1.102 --type ban
```

CrowdSec Firewall IP Blocking Command Execution

6. Case Management (TheHive)

   A case was created automatically to track the incident.

```
Case Title: Phishing IP Detected
Severity: High
IP Address: 192.168.1.102
Status: Open
Description: Suspicious IP detected from phishing alert and blocked
automatically.
```

TheHive case Created for phishing incident

7. Playbook Summary

   A SOAR playbook was developed to automate phishing incident response. The playbook detects alerts from Wazuh, checks IP reputation, blocks malicious IPs using CrowdSec, and creates a case in TheHive. This automation reduces response time, minimizes manual effort, and ensures consistent and efficient handling of security incidents.

Post-Incident Analysis and Metrics

1. Objective
   The objective of this task is to perform root cause analysis (RCA) on a security incident, identify contributing factors, and evaluate SOC performance using key metrics such as MTTD and MTTR.

2. Tools Used
   - Google Sheets
   - Draw.io

3. Root Cause Analysis
   A phishing incident was analysed using the **5 Whys technique** to identify the root cause.
   - User clicked on a malicious link
   - Weak email filtering allowed the phishing email
   - Lack of advanced detection mechanisms
   - Insufficient security tools
   - Poor security policy implementation

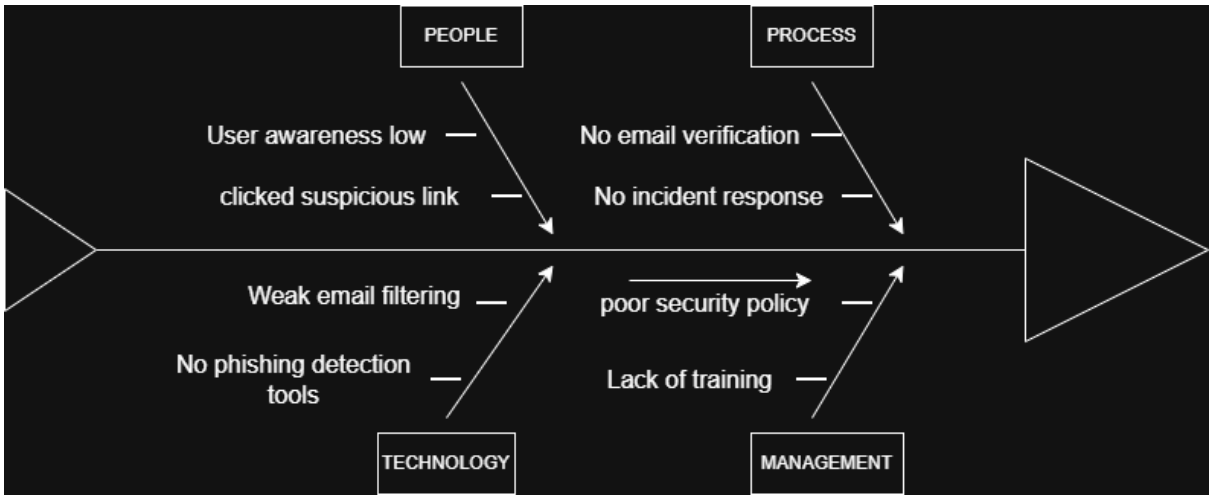
	A	B
1	**QUESTION**	**ANSWER**
2	Why was email opened?	User clicked malicious link
3	Why was link clicked?	Weak email filtering
4	Why was filtering weak?	No advanced spam/phishing detection
5	Why no detection system?	Lack of security tools
6	Why lack of tools?	Poor security policy implementation

Root Cause Analysis Using 5 Whys Method

4. Fishbone Diagram (Cause Analysis)

   A Fishbone Diagram was created to categorize the causes of the incident into:

   - People
   - Process
   - Technology



Fishbone Diagram for incident Cause Analysis

This diagram visually represents different contributing factors leading to the incident, helping in better understanding and prevention planning.

5. SOC Metrics Calculation

   Security metrics were calculated to evaluate response performance.

   Metrics:

   - **MTTD (Mean Time to Detect):** 2 Hours
   - **MTTR (Mean Time to Respond):** 4 Hours
   - **False positive rate 5%**

	A	B
1	**METRIC**	**VALUE**
2	MTTD	2 Hours
3	MTTR	4 Hours
4	False Positive rate	5%

SOC Metrics – MTTD, MTTR, and False Positive Rate

This screenshot shows the calculated SOC performance metrics including MTTD, MTTR, and false positive rate. The detection time was 2 hours and response time was 4 hours, while the false positive rate was 5%, indicating relatively accurate alert detection.

6. Summary

A comprehensive post-incident analysis was conducted to evaluate a simulated phishing attack and identify its root causes. The 5 Whys technique was applied to systematically investigate the incident, revealing that the primary cause was weak email filtering and lack of advanced detection mechanisms. Further analysis indicated that insufficient security tools and poor policy implementation also contributed to the success of the attack. This highlights the importance of strong security infrastructure and user awareness in preventing such incidents.In addition to root cause analysis, key Security Operations Center (SOC) performance metrics were calculated to assess the effectiveness of detection and response processes. The Mean Time to Detect (MTTD) was recorded as 2 hours, while the Mean Time to Respond (MTTR) was 4 hours, indicating a moderate response efficiency. Furthermore, the false positive rate was found to be 5%, suggesting that most alerts were accurate with minimal unnecessary triggers.

Alert triage and Automation

1. Objective
   The objective of this task is to analyze and triage security alerts, validate them using threat intelligence tools, and automate the verification process to improve response efficiency.

2. Tools Used
   - Wazuh
   - VirusTotal
   - TheHive

3. Alert Triage
   A security alert was analyzed to determine its severity and validity.
   Alert Details:
   - **Alert ID:** 005
   - **Description:** Suspicious File Download
   - **Source IP:** 192.168.1.102
   - **Priority:** High
   - **Status:** Open

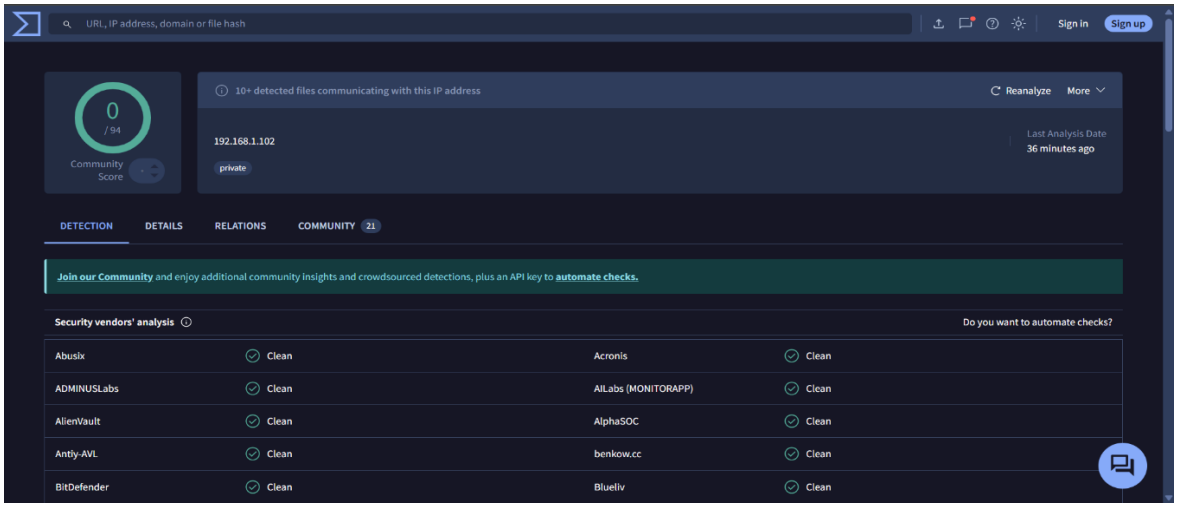
   The alert was classified as high priority due to potential malicious activity.

Alert ID	Description	Source IP	Priority	Status
001	Samba exploit attempt	192.168.1.100	High	Confirmed

Alert Triage Details in Wazuh

4. Automated Validation using Threat Intelligence
   The alert was validated using threat intelligence tools.

   Process:
   - File/IP checked using VirusTotal
   - Reputation analysis performed
   - Automated validation configured



Automated Validation using Threat Intelligence

5. Automation using TheHive

Automation was configured to integrate TheHive with threat intelligence tools.

Actions:

- Automatic hash/IP checking
- Alert enrichment
- Case creation

6. Finding Summary

- Alert successfully identified and analyzed
- Threat intelligence validation performed
- No strong malicious indicators found
- Automation improved efficiency of triage process

7. Summary

An alert triage process was performed on a suspicious file download event using Wazuh. The alert was analysed and validated using VirusTotal, which showed no strong malicious indicators. Automation using TheHive helped streamline the validation process, reducing manual effort and improving the efficiency of security operations.

Evidence Analysis

1. Objective
   The objective of this task is to analyse collected forensic evidence, identify suspicious activities, and maintain proper chain-of-custody to ensure integrity and reliability of the investigation.

2. Tools Used
   - Velociraptor
   - FTK Imager

3. Evidence Analysis
   Evidence was collected and analysed from system logs and network activity.
   ❖ Activity Performed:
     - Analysed network connections using queries
     - Identified potential suspicious connections
     - Reviewed system-level evidence

4. Chain of Custody
   To ensure the integrity of evidence, a chain-of-custody record was maintained.
   ❖ Details Recorded:
     - Evidence item
     - Description
     - Collected by
     - Date
     - Hash value

Item	Description	Collected By	Date	Hash Value
Network Log	Server-Z Log	SOC Analyst	2025-08-18	3f5d8c2a9b1e4f6d7c8a9b0e1f2d3c4b5a6e7f8d

Evidence Chain-of-custody Record

5. Finding Summary
   - Evidence successfully collected and analysed
   - Chain-of-custody maintained properly
   - Data integrity ensured using hash values
   - Investigation process remained reliable

6. Summary
   Evidence analysis was performed using Velociraptor to examine system and network activity. A proper chain-of-custody record was maintained to ensure data integrity and traceability. This process helps in reliable forensic investigation and supports accurate identification of suspicious activities during a security incident.

Adversary Emulation Practice

1. Objective
   The objective of this task is to simulate real-world attacker techniques using adversary emulation tools and evaluate the detection capabilities of the security monitoring system.

2. Tools Used
   - MITRE Caldera
   - Wazuh

3. Adversary Simulation
   An adversary emulation was performed to simulate a phishing attack.
   - Technique Used:
     - T1566- Phishing

4. Detection using Wazuh
   The simulated attack was monitored using Wazuh to verify detection.

- ❖ Observation
  - Attack was successfully detected
  - Alert generated by the system

Timestamp	TTP	Detection Status	Notes
2025-08-18 17:00:00	T1566	Detected	Phishing email blocked

Adversary Emulation Detection Results

This screenshot shows the detection of the simulated phishing attack using Wazuh. The system successfully identified the T1566 technique and generated an alert, confirming the effectiveness of the detection mechanism.

5. Finding Summary
   - Adversary simulation executed successfully
   - Detection system identified the attack
   - Alerts generated correctly
   - Security monitoring proved effective

6. Emulation Report
   An adversary emulation exercise was conducted using MITRE Caldera to simulate a phishing attack based on MITRE ATT&CK technique T1566. The attack was successfully detected by Wazuh, which generated alerts indicating suspicious activity. The results demonstrate that the detection system is capable of identifying phishing-based threats effectively. However, continuous monitoring and improvement are necessary to handle more advanced attack techniques. This exercise highlights the importance of adversary emulation in testing and validating security defenses, ensuring that systems are prepared to detect and respond to real-world cyber threats.

Security Metrics and Executive Reporting

1. Objective
   The objective of this task is to calculate key security metrics, analyse SOC performance, and present findings in a structured executive report for better decision-making.

2. Tools Used
   - Elastic Security
   - Google Sheets
   - Google Docs

3. Metrics Calculation
   Key SOC performance metrics were calculated:
   - **MTTD (Mean Time to Detect):** 2 Hours
   - **MTTR (Mean Time to Respond):** 4 Hours
   - **False Positive Rate:** 5%
   - **Dwell Time:** 2 Hours

	A	B
1	**METRIC**	**VALUE**
2	MTTD	2 Hours
3	MTTR	4 Hours
4	False Positive rate	5%

SOC Performance Metrics Dashboard

This screenshot shows the calculated SOC performance metrics including detection time, response time, and false positive rate.

These metrics help in evaluating the efficiency and accuracy of the security operations.

4. Dwell Time Analysis
   Dwell time represents the duration between attack occurrence and detection.
   Observation:
   - Incident Start Time: 10:00
   - Detection Time: 12:00
   - **Dwell Time:** 2 Hours

	A	B	C
1	**INCIDENT START**	**DETECTION TIME**	**DWELL TIME**
2	10:00	12:00	2 Hours

Dwell Time Calculation

This screenshot shows the calculation of dwell time, which is the time taken to detect an incident after it occurs. A dwell time of 2 hours indicates moderate detection efficiency.

5. Summary
   The security analysis was conducted to evaluate the performance of the Security Operations Center (SOC) using key metrics such as Mean Time to Detect (MTTD), Mean Time to Respond (MTTR), false positive rate, and dwell time. The results indicated that the system detected threats within 2 hours and responded within 4 hours, demonstrating a moderate level of efficiency. The false positive rate was measured at 5%, indicating a relatively accurate alerting system with minimal unnecessary alerts. Additionally, the dwell time of 2 hours suggests that threats were identified within a reasonable timeframe. The analysis highlights the effectiveness of the current security infrastructure while also identifying areas for improvement.

Conclusion

This project successfully demonstrated the design and implementation of a comprehensive Security Operations Center (SOC) workflow by integrating both theoretical knowledge and practical cybersecurity operations. The project covered multiple critical domains including threat hunting, SOAR automation, alert triage, post-incident analysis, adversary emulation, evidence handling, and security metrics reporting. Each phase was executed using industry-relevant tools and methodologies, providing a hands-on understanding of real-world security operations. During the threat hunting phase, proactive techniques were applied to identify potential threats based on hypotheses and attacker behaviour patterns. The use of threat intelligence platforms enhanced the ability to detect suspicious activities and understand attacker tactics such as misuse of valid accounts. In the SOAR automation phase, automated playbooks were developed to handle phishing incidents efficiently. This significantly reduced manual effort and ensured faster and more consistent incident response. The integration of different tools allowed seamless orchestration and improved workflow efficiency. The project also included detailed post-incident analysis using techniques such as the 5 Whys method and performance metrics evaluation. This helped in identifying root causes of incidents and improving future response strategies. Adversary emulation using tools like MITRE Caldera enabled the simulation of real-world attack scenarios, which validated the effectiveness of detection mechanisms. Additionally, evidence analysis and chain-of-custody practices ensured proper handling and integrity of forensic data during investigations.